

Dhyani Soni

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EDUCATION

NEW JERSEY INSTITUTE OF TECHNOLOGY

B.S. in Computer Science & AI Minor, Albert Dorman Honors College

Expected May 2027

Newark, NJ

GPA: 3.62 | Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Artificial Intelligence, Data Science, Linear Algebra, Probability & Statistics, AI-Assisted Software Engineering, Corporate Finance

- **Campus Involvement:** Lyra STEM Curriculum Manager, Kids Who Code Treasurer & Society of Hispanic Professional Engineers (SHPE) Member

SKILLS

- **Languages:** Python, SQL, Java, JavaScript, C/C++, HTML/CSS
- **Frameworks & Libraries:** Spring Boot, React, FastAPI, Streamlit, TensorFlow, YOLO, Pytorch, FAISS Tools, Tailwind CSS, Flask, NumPy, Pandas, Scikit-learn, Matplotlib
- **Platforms:** VS Code, AWS (S3, RDS), Docker, MySQL, Jupyter Notebook, Git, Tableau, Postman, Anaconda, GitHub Copilot, MS Office Suite, Figma

PROFESSIONAL EXPERIENCE

Artificial Intelligence & Machine Learning Fellow

Mar 2026 - Present

Break Through Tech @ Cornell Tech

- Selected nationally from **4,000+ applicants** for a rigorous year long **AI Fellowship at Cornell Tech**
- Developing **hands-on machine learning expertise** through an accelerated Python bootcamp and advanced coursework in data wrangling, predictive modeling, and performance evaluation, applying **Pandas, Scikit-Learn, and Matplotlib**
- Scheduled to lead an **AIStudio project** with an **industry sponsor** (Aug2026–Mar2027), managing a cross-functional team to ship a **deployment-ready ML product** and deliver final outcomes to **senior executive leadership**

Artificial Intelligence & Machine Learning Research Intern

Sept 2025 - May 2026

CNALab @ New Jersey Institute of Technology

- Won **1st Place** at NJIT's Honors Interdisciplinary Research Forum (**40+ competitors**) by delivering an interdisciplinary research presentation that clearly connected methods to **real-world impact**; also selected as a **Finalist** for the Undergraduate Research and Innovation Program.
- Developed an AI-powered rehabilitation tool for **patient mobility tracking** in aquatic therapy by implementing pose estimation with **OpenCV** and **Python**; built a custom preprocessing pipeline with **image enhancement**
- **Improved keypoint detection by 35%** and reducing labeling errors in the training dataset, increasing joint torque and acceleration analysis accuracy underwater
- Collaborated **cross-functionally** with clinical researchers to align model outputs

UX Research Assistant

May 2024 - Dec 2024

Department of Informatics @ New Jersey Institute of Technology

- Led qualitative UX research for **inclusive interface design**; conducted and analyzed 25+ interviews with older adults to uncover **accessibility gaps in digital safety**
- Converted behavioral insights into actionable recommendations via UX **storyboards** and **strategy reports**
- Applied **market research techniques** like user personas, segmentation analysis, and journey mapping to inform CX and product strategies in customer-facing roles, influencing inclusive **UI changes** for campus-wide platforms

PROJECTS

ADPVerse | Python, SQLite, ElevenLabs, Presage, React

Feb 2026

- Won **Best AI Analytics Track at HackHers Hackathon (ADP Challenge)** by architecting a real-time multimodal sentiment analysis pipeline for sales pitch evaluation at **ADP**
- Integrated Presage (emotion detection), ElevenLabs (speech-to-text), and a **timestamped database** to structure client interaction data
- Leveraged **Gemini** for **contextual mood analysis**, notifying reps of negative engagement and **suggest** optimized **high-impact pitch strategies**.

AgentWeave | Python, FastAPI, YOLOv5, FAISS, CLIP, React, Firebase

July 2025

- Engineered a scalable AI-driven clothing recommendation tool using YOLOv5 and CLIP to detect users and generate outfit matches based on visual similarity
- **Deployed FAISS vector** search for real-time outfit retrieval, and built a **swipe-based user interface in React** to collect behavioral feedback and write preference data directly to Firebase for usage analytics
- Clustering user interactions with embedding vectors to dynamically refine model accuracy and **personalize recommendations**